

Medicare Advantage Quality Improvement Dashboard

Independent portfolio case study using synthetic healthcare quality records to identify improvement priorities, compare outreach results, and monitor data reliability.

Business question

Where should a quality improvement team focus follow-up when measure performance varies across providers and regions?

Scope and method

Created 8,000 member-measure records representing 3,075 synthetic members assessed during January-June 2026. Used SQL to calculate eligible-denominator rates, target shortfalls, monthly cohort trends and outreach results. Kept a separate staging dataset for ten audit rules.

Selected findings

Overall compliance is 70.9% (4,173/5,884), with 1,711 open gaps. Colorectal screening and post-hospital follow-up account for 308 of 531 additional closures needed to meet all demonstration targets. Provider Group C is at 61.2% versus Group A at 77.5%. A further 545 open gaps have no outreach attempt.

Recommended actions

Prioritize the two highest-shortfall measures, review Provider Group C and the South region within each measure, work the unattempted backlog, and test coordinator escalation for unresolved gaps. Validate source records before each refresh.

Delivered work

SQL queries and a ready-to-query SQLite database, an Excel analysis workbook, a working four-page browser dashboard, a data dictionary, seven findings and five recommendations. Power BI deliverables include DAX, model relationships, a theme and exact visual mappings for all four pages. Native Power BI Desktop implementation remains the final build step.

Demonstrated value

Shows practical analyst skills in metric definition, SQL, reporting, data validation and translating findings into operational priorities. The 531-closure opportunity is an arithmetic planning estimate, not a realized healthcare or financial result.

Interpretation

Targets and eligibility rules are demonstration assumptions. Monthly cohorts differ. Channel results are descriptive. This project does not calculate official CMS Stars or certified HEDIS measures.

This project uses entirely synthetic data created for demonstration purposes. No PHI, PII, patient records, or proprietary employer data are included.